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Topics in Probability Theory and Stochastic Processes
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Markov Chains

Rating

Mathematically Mature: may contain mathematics beyond calculus with proofs.

Section Starter Question

Consider the magical Land of Oz where the weather follows a pattern. If it is raining today, then there is a 50% chance of raining again tomorrow, and a 25% chance of either having a nice day or a snowy day tomorrow. Similarly if it is snowing today, there is a 50% chance of having snow tomorrow, and a 25% chance of either having a nice day, or a rainy day tomorrow. Also the land of Oz never has two nice days in a row, with equal chance of rain or snow the day after a nice day. What is the chance it is nice in two days given it is rainy today?

Key Concepts

1. A **stochastic process** is, in the simplest terms, a random function. A specific outcome of the process must be specified in terms of the probability of it happening.

2. A discrete time discrete state space stochastic process is a **Markov chain** if $\mathbb{P}[X_{n+1} \mid X_n, X_{n-1}, \ldots, X_1] = \mathbb{P}[X_{n+1} \mid X_n]$. That is, the conditional probability depends only on the immediately preceding state, not the history.

3. Denote the probability of making the transition from state x_i to x_j at time n as:
$$p_{ij}(n) = \mathbb{P}[O_{n+1} = x_j \mid O_n = x_i].$$
If the probability of a state transition in a Markov process does not depend on n, then it is a **time-homogeneous Markov chain**. If the state space is finite, say $|\mathcal{X}| = k$, then the p_{ij} are the entries of an $k \times k$ matrix P called the **transition probability matrix**.

4. The transition probability for multiple time steps is the matrix P^n, and this is known as (the finite Markov Chain version of) the *Chapman-Kolmogorov Theorem*.

5. The Fundamental Theorem of Markov Chains says: Let $\mathcal{X}$ be a finite state space and let P_{ij} define a Markov chain on x_i. If there is an integer n_0 such that $(P^{n_0})_{ij} > 0$ for all $n > n_0$, then P has a unique stationary distribution π, and as $n \to \infty$, $(P^n)_{ij} \to \pi_j$ for each i, j.

6. **Absorbing states** are states that once entered are never left.

7. The limiting matrix can be found by putting the matrix in **canonical form**, then finding the **fundamental matrix**, and then multiplying blocks from the canonical matrix.

Vocabulary

1. A **stochastic process** is a random function on a domain.

2. The outcome of a **discrete time discrete state space process** is one of several states at a sequence of times. Along a specific sample, as time progresses the observed sequence of states from the set $\mathcal{X}$ is a **sample path**.

3. A **discrete time Markov process** is a stochastic process having a fixed probability that it will be in state x_j at time $n+1$ when the process is in state x_i at time n, with no dependence on the prior history.

4. If the probability of a state transition in a Markov process does not depend on n, it is a **time-homogeneous Markov chain**.

5. If the values in each the rows of a matrix sum to 1, it is a **stochastic matrix**.

6. A **Markov chain** is a discrete time discrete state space defined by its transition probability matrix P with entries

$$p_{ij}(n) = \mathbb{P}\left[X_{n+1} = x_j \mid X_n = x_i\right].$$

The entries must satisfy $P_{ij} \geq 0$ and $\sum_{j=1}^{k} P_{ij} = 1$ for each $i = 1, \ldots, k$, so that each row is a probability distribution on X.

7. States which once entered are never left are **absorbing states**.

8. Given a Markov chain defined by transition matrix P with u absorbing states and v non-absorbing states where $u + v = k$, then the matrix has the **transition probability matrix canonical form**

$$P = \begin{pmatrix} I & 0 \\ R & Q \end{pmatrix}$$

where I is the $u \times u$ identity matrix, and 0 is the $u \times v$ matrix of zeros. The matrix $N = (I - Q)^{-1}$ is called the **fundamental matrix**.

Mathematical Ideas

Introduction

This section is a survey of basic concepts and facts about Markov processes and Markov chains. Detailed treatments are in [1, 2, **?**, 3, 4, 5, 6, 7], among other standard references.

Begin by understanding where Markov chains fit in the larger landscape of mathematics.

Definition. A **stochastic process** is a random function on a domain. The domain can be discrete or continuous. The domain is often interpreted as time. Stochastic processes are in contrast to deterministic processes. A specific outcome of the process at some time must be specified in terms of the probability of it happening.

Consider the following simple example:

$$f(t) = 2t$$

$$g(t) = \begin{cases} 2t & \text{with probability } 0.98 \\ 3t & \text{with probability } 0.02. \end{cases}$$

For $f(2)$ the outcome will always be 4, it is deterministic. However for $g(2)$ the most likely outcome is 4 but this will fail to be true for about 2% of the trials.

Definition. The outcome of a **discrete time discrete state space stochastic process** is one of several discrete states at a sequence of times. For the purposes of this introduction call these states $\mathcal{X} = \{x_1, x_2, \ldots, x_i, \ldots, \}$. Sometimes the state space $\mathcal{X}$ is infinite, but often the state space is finite. At time n, guided by some information, the process acts in a probabilistic manner and outputs some x_i. Thus along a specific sample, as time progresses an observed sequence of states from the set $\mathcal{X}$ is a **sample path**, denoted $\mathbf{O} = \{O_1, O_2, \ldots\}$.

Definition. A **discrete time Markov process** is a stochastic process having a fixed probability that it will be in state x_j at time $n+1$ when the process is in state x_i at time n, with no dependence on the prior history. Symbolically,

$$\mathbb{P}\left[X_{t_{n+1}} \mid X_{t_n} = O_n, X_{t_{n-1}} = O_{n-1}, \ldots X_{t_1} = O_1, X_{t_0} = O_0\right] = \mathbb{P}\left[X_{t_{n+1}} \mid X_{t_n} = O_n\right]$$

Denote the probability of making the transition from state x_i to x_j at time n as:

$$P_{ij}(n) = \mathbb{P}\left[O_{n+1} = x_j \mid O_n = x_i\right].$$

As a contrast, consider modeling the daily closing price of a given company's stock. At time n, with a wealth of available information: the past prices of the stock, the current market conditions, the weather forecast for the next few days, even the price of tea in China. Instead, modeling the daily closing price of a stock as a discrete time Markov process, at time $n+1$ the *only* information that would affect the process would be the observed stock price at time n.

Definition. If the probability of a state transition in a Markov process does not depend on n, then it is a **time-homogeneous Markov chain**. This means $P_{ij}(n) = P_{ij}(m)$ for all n, m. That is, the transition probabilities from x_i to x_j are time invariant with fixed P_{ij}, called the transition probabilities from x_i to x_j. If the state space is finite, say $|\mathcal{X}| = k$, then the P_{ij} are the entries of an $k \times k$ matrix P called the transition probability matrix.

Example of a Markov Chains

The following is a classic example from [3], repeated in [2] and many other locations. The example puts the concepts in context as well as introducing some other ideas.

Example. Consider the magical Land of Oz where the weather follows a pattern. If it is raining today, then there is a 50% chance of raining again tomorrow, and a 25% chance of either having a nice day or a snowy day tomorrow. Similarly if it is snowing, there is a 50% chance of again having snow, and a 25% chance of either having a nice day, or a rainy day. The land of Oz never has two nice days in a row, and Oz equally has rain and snow the day after a nice day.

First label the states conveniently as R, N, S, instead of x_1, x_2, x_3. With the information above, define $P_{RR} = 1/2$, $P_{RN} = 1/4$ and so on. Writing out every possible transition gives 3^2 distinct possibilities. With k states, the process has k^2 transitions with the probability P_{ij} of going from x_i to x_j.

Use matrices to represent the transition probabilities for the Markov chain. In the example, the transition matrix is (the row and column labels of R, N and S are for convenience only and are not usually included in the display of the matrix).

$$P = \begin{array}{c} \\ R \\ N \\ S \end{array} \begin{array}{c} \begin{array}{ccc} R & N & S \end{array} \\ \begin{pmatrix} 1/2 & 1/4 & 1/4 \\ 1/2 & 0 & 1/2 \\ 1/4 & 1/4 & 1/2 \end{pmatrix} \end{array}.$$

Note that all the rows sum to 1, thus this is a **stochastic matrix**.

How to calculate the chance it is nice in two days given it is rainy today? Use the Law of Total Probability to calculate this chance. This considers every possible weather combination with their respective conditional probabilities. See Figure 1 for a diagram of the combination process. Mixing the $R = x_1$, $N = x_2$, $S = x_3$ notation, this example has

$$\Pr[O_{n+2} = N | O_n = R] = \sum_{j=1}^{3} P_{1j} P_{j2}.$$

For two days, this is not too tedious, but what about the probability it will be nice a week from today given it is rainy today?

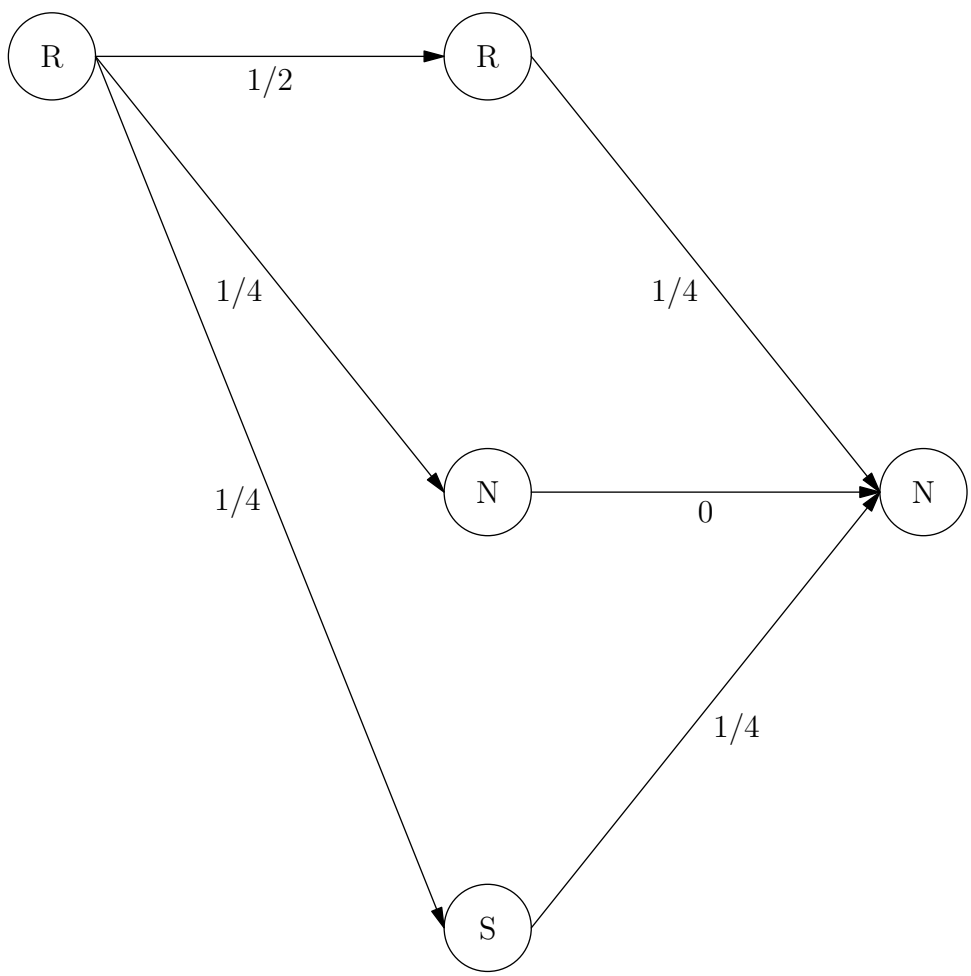


Figure 1: A schematic diagram of the calculation of the two-step transition probability.

Now matrix notation is convenient. Notice that the probability above of going from R to N two days hence is the $(1,2)$ entry in the matrix product of P with itself. More generally, the transition probability through n steps is the matrix P^n, and this is known as (the finite Markov Chain version of) the *Chapman-Kolmogorov Theorem*. Thus consider the following matrices from the example:

$$P^2 = \begin{array}{c} \\ R \\ N \\ S \end{array} \begin{array}{c} \begin{array}{ccc} R & N & S \end{array} \\ \begin{pmatrix} 7/16 & 3/16 & 6/16 \\ 6/16 & 4/16 & 6/16 \\ 6/16 & 3/16 & 7/16 \end{pmatrix} \end{array}$$

and

$$P^7 = \begin{array}{c} \\ R \\ N \\ S \end{array} \begin{array}{c} \begin{array}{ccc} R & N & S \end{array} \\ \begin{pmatrix} 0.400024 & 0.200012 & 0.399963 \\ 0.400024 & 0.199951 & 0.400024 \\ 0.399963 & 0.200012 & 0.400024 \end{pmatrix} \end{array}.$$

Hence if it is raining today, then the chance it is nice in 2 and 7 days is $3/16$ and 0.200012 respectively.

Formal Definitions

Let the state space be a finite set $\mathcal{X} = \{x_1, x_2, \ldots, x_k\}$.

Definition. A **Markov chain** is a discrete time discrete state space stochastic process defined by its transition probability matrix P with entries

$$P_{ij}(n) = \mathbb{P}\left[X_n = x_j \mid X_{n-1} = x_i\right].$$

The entries must satisfy $P_{ij} \geq 0$ and $\sum_{j=1}^{k} P_{ij} = 1$ for each $i = 1, \ldots, k$, so that each row is a probability distribution on X.

This means that the Markov chain is a random walk among the states with dependence only on the current state. If x_i is the current state, choose x_j with probability P_{ij}; from x_j choose x_l with probability P_{jl}, and so on. The sequence of outcomes $X_0 = x_i, X_1 = x_j, X_2 = x_l \ldots$ is a **sample path** of the Markov chain.

In terms of conditional probabilities,

$$\mathbb{P}\left[X_1 = x_j \mid X_0 = x_i\right] = P_{ij}$$
$$\mathbb{P}\left[X_2 = x_l, X_1 = x_j \mid X_0 = x_i\right] = P_{ij}P_{jl}.$$

Then combining over all the intermediate possibilities

$$\mathbb{P}\left[X_2 = x_l \mid X_0 = x_i\right] = \sum_{j=1}^{k} P_{ij} P_{jl}$$

which is matrix multiplication. This is the *Chapman-Kolmogorov equation* for finite Markov chains. Extending this observation by induction

$$\mathbb{P}\left[X_n = l \mid X_0 = i\right] = (P^n)_{il}$$

the i, l entry of the nth power of P.

A Markov chain has a **stationary distribution** if there is a vector π satisfying

$$\sum_{i=1}^{k} \pi_i P_{ij} = \pi_j$$

with $\pi_i > 0$, and $\sum_{i=1}^{k} \pi_i = 1$. Thus π is a left eigenvector of P with eigenvalue 1. Alternative terminology is that the Markov chain is **stable on the distribution** π and that π is the **stable distribution** for the Markov chain. The probabilistic interpretation of the left eigenvector equation is "pick x_i from $\mathcal{X}$ with probability π_i and take a step with probability P_{ij}, the probability of being at x_j is π_j". Thus π is stationary for the evolution of the Markov chain.

The precise definitions of the conditions on a Markov chain for the following Fundamental Theorem to hold are in the next section "Classes of States and Stationary Distributions". The proof of the Fundamental Theorem is also in the next section "Classes of States and Stationary Distributions".

Theorem 1 (Fundamental Theorem of Markov Chains)**.** *For an irreducible, positive recurrent and aperiodic Markov chain* $\lim_{n\to\infty}(P^n)_{ij}$ *exists and is independent of* i*. Furthermore, letting*

$$\pi_j = \lim_{n\to\infty} (P^n)_{ij}$$

then π_j *is the unique non-negative solution of*

$$\sum_i \pi_i P_{ij} = \pi_j,$$

$$\sum_i \pi_i = 1.$$

An irreducible, positive recurrent and aperiodic Markov chain with transition matrix P has the following property: As $n \to \infty$, the matrix P^n approaches a limiting matrix P^∞, where P^∞ consists of n copies of the same row vector. This matrix has different names in different texts, two common names are the "limit matrix", or the matrix of "limiting probabilities."

The probabilistic content of the Fundamental Theorem is that from any starting state x_i, for large n the nth step of a sample path of the Markov chain has a probability close to π_j of being at x_j if n is large. Later sections address the question of what "probability close to π_j" means and how large n must be for this probability relationship to hold.

Example. For the weather in Oz example, consider P^7 further:

$$P^7 = \begin{pmatrix} & R & N & S \\ R & 0.400024 & 0.200012 & 0.399963 \\ N & 0.400024 & 0.199951 & 0.400024 \\ S & 0.399963 & 0.200012 & 0.400024 \end{pmatrix}.$$

Note that the chance of being in any of the three states after seven days is almost independent of the initial state. This means the weather in seven days is almost completely independent of what it is today. Considering P^n for larger values of n, the row vectors approach $\pi = (2/5, 1/5, 2/5)$.

This works for the example, but it will not work for every matrix. Consider:

$$Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This matrix describes a two-state system switching states at each time step. Thus $Q^2 = I$, $Q^3 = Q$, $Q^4 = I$ and so on.

Absorbing States

Up to this point the only example has been an irreducible, positive recurrent and aperiodic Markov chains. Now consider the following transition matrix:

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Note that if the process starts in state x_1 or x_5, it can't leave these states. Similarly, if the process starts in any other state and "arrives" in states x_1 or x_5 at time N, then it stays in those states for all steps $n \geq N$. Call these **absorbing states**, that is, states which once entered are never left.

The transition matrix P represents the random walk on x_0, x_1, x_2, x_3, x_4 that stops at x_0 or x_4. Markov chains with absorbing states are *not* irreducible. Therefore the result above does not apply for a "limit" matrix, so introduce the **transition probability matrix canonical form**. Given a Markov chain defined by transition matrix P with u absorbing states and v non-absorbing states where $u + v = k$, , then the matrix has the canonical form:

$$P = \begin{pmatrix} I & 0 \\ R & Q \end{pmatrix}$$

where I is the $u \times u$ identity matrix, and 0 is the $u \times v$ matrix of zeros. The random walk matrix from above has the canonical form:

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 1/2 & 0 & 1/2 & 0 \end{pmatrix}.$$

The matrix $N = (I - Q)^{-1}$ is called the **fundamental matrix**. The entries of the fundamental matrix have the following interpretation, N_{ij} is the expected number of times the chain is expected to be at state x_j before absorption, given it started in state x_i. So in the random walk example:

$$N = (I - Q)^{-1} = \begin{pmatrix} 3/2 & 1 & 1/2 \\ 1 & 2 & 1 \\ 1/2 & 1 & 3/2 \end{pmatrix}.$$

Letting $\mathbf{1}$ be the vector of all ones, $t = N\mathbf{1}$ is the expected number of steps before absorption in one of the absorbing states. Define $B = NR$, so B_{ij} is the probability that starting in the ith non-absorbing state, the process ends up in the jth absorbing state. In the example:

$$B = (I - Q)^{-1}R = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \\ 1/4 & 3/4 \end{pmatrix}.$$

This allows us to comment about P^n for large values of n. As $n \to \infty$, P^n approaches the matrixP^∞ defined as:

$$P^\infty = \begin{pmatrix} I & 0 \\ B & 0 \end{pmatrix}.$$

Section Ending Answer

Use the Law of Total Probability to calculate this probability. This considers every possible weather combination with their respective conditional probabilities.

$$\mathbb{P}\left[O_2 = N \mid O_0 = R\right] = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{4} \cdot 0 + \frac{1}{4} \cdot \frac{1}{4} = \frac{3}{16}.$$

Sources

This section is adapted from: Notes prepared by LT Grant and used with permission. Pieces of this section are adapted from: *Finite Markov Chains*, Kemeny and Snell, 1960, and from *Random Walks and Electric Networks*, Doyle and Snell, 1984, as well as *A First Course in Stochastic Processes* by S. Karlin and H. Taylor, Chapter 2, 1975.

Algorithms, Scripts, Simulations

Algorithm

Markov Chain Simulation

```
Comment Post: Information about a simple Markov chain
Load Markov chain library
Set state names
Set transition probability matrix
Set start state
Set up Markov chain with the library
Set an example length
Create a sample path of example length
Create a second sample path of example length
Set a long path length, and a transient time
Create a long sample path
Slice the long sample path from the transient time to the end
In the slice count the appearance of each state
Store in an empirical array
Compute the theoretical stable array
return Empirical stable distribution
return Theoretical stable distribution
```

Scripts

R R script for markovchain.

```r
library(markovchain)

stateNames <- c("R", "N", "S")
transMatrix <- matrix( c(1/2, 1/4, 1/4, 1/2, 0, 1/2, 1/4, 1/4, 1/2),
                       nrow=3, byrow=TRUE)
startState <- "N"

weatherOz <- new("markovchain", transitionMatrix= transMatrix,
                 states=stateNames, name="Weather_in_Oz")
print( weatherOz )

print( summary(weatherOz) )
```

```r

pathLength <- 10
weatherPath <- rmarkovchain(n=pathLength,
                            object = weatherOz, t0 = startState)
cat("A sample starting from N: ", weatherPath, "\n")
weatherPath <- rmarkovchain(n=pathLength,
                            object = weatherOz, t0 = startState)
cat("Another sample starting from N: ", weatherPath, "\n")

largeN <- 1000; startStable <- 100
weatherPath <- rmarkovchain(n=largeN,
                            object = weatherOz, t0 = startState)
stablePattern <-  weatherPath[startStable:largeN]
empiricalStable <- c( mean( stablePattern == "R" ),
                      mean( stablePattern == "N" ),
                      mean( stablePattern == "S" ))
theorStable <- steadyStates(weatherOz)
cat("Empirical stable distribution: ",  empiricalStable, "\n")
cat("Theoretical stable distribution: ", theorStable, "\n")
```

Problems to Work for Understanding

1: Students' progress through a (two-year) Associate Degree program is not always a two year process. A student taking first year courses has a 15% chance of dropping out, a 25% chance of repeating the first year, and a 60% chance of moving on to the second year. A student taking second year material has a 10% chance of dropping out, a 20% of repeating the second year, and a 70% chance of graduating.

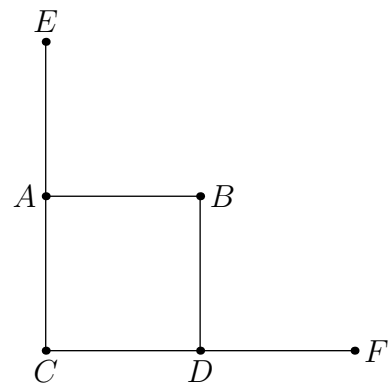


Figure 2: Grid for the random walk.

(a) How many states are in this model?

(b) Which states are absorbing? Which are transient?

(c) What is the chance a student entering the program this year will graduate "on time"? What is the chance a student will graduate in 3 years?

(d) What is the chance that a student will graduate eventually? What is the chance the student will drop out?

(e) What is the expected time until a student graduates?

2: Consider the grid in Figure 2 upon which a random walker is stuck. If she reaches states E and F she will stop, otherwise she will keep walking. At any of the other states, she is equally likely to choose any of the possible neighboring states.

(a) What is the transition matrix?

(b) What are the absorbing states? The transient states?

(c) What is the canonical form of this matrix? The fundamental form?

(d) Starting at state A, on average, how many steps will the walker take before reaching states E or F?

(e) Starting at state A, what is the probability the walker will end at the state F? Starting at D what is the probability the walker ends at state E? Discuss these two answers.

3: The walker returns a year later to the same grid, only to find that there now exists a road between B and E.

(a) Without any calculations, how would you expect this to change the absorption probabilities of state B?

(b) Without any calculations, would you expect the same results as you found in part (e) above?

(c) Find the new transition matrix.

(d) Calculate the actual probabilities for part (a) and (b) and comment on your results.

4: Determine the transition probability matrix for the following Markov Chain: N black balls and N white balls are placed in two urns so that each contains N balls. At each step one ball is selected at random from each urn and the two balls interchange urns. The state of the system is the number of white balls in the first urn. What are the absorbing states, transient states, recurrent states, if any? Is this Markov chain ergodic?

5: Determine the transition probability matrix for the following Markov Chain: Consider two urns A and B with a total of N balls. An experiment is performed in which a ball is selected at random (all selections equally likely) at time n, $n = 1, 2, \ldots$ from among the totality of N balls. Then an urn is selected at random, urn A with probability p and urn B with probability $q = 1-p$ and the ball previously drawn is placed in this urn. The state of the system is the number of balls in the urn A. What are the absorbing states, transient states, recurrent states, if any? Is this Markov chain ergodic?

6: Determine the transition probability matrix for the following Markov Chain: Consider two urns A and B with a total of N balls. An experiment is performed in which an urn is selected at random in proportion to its contents (i.e. if urn A has exactly k balls, then it is selected with probability k/N. The a ball is selected from A with probability p and from urn B with probability $q = 1 - p$ and placed in the previously chosen urn. The state of the system is the number of balls in the urn A. What are the absorbing states, transient states, recurrent states, if any? Is this Markov chain ergodic?

7: Your baby is learning to walk. The baby begins by holding onto a couch. Whenever she is next to the couch, there is a 25 percent chance that she will take a step forward and a 75 percent chance that she will stay

clutching the couch. If the baby is one or more steps away from the couch, there's a 25 percent chance that she will take a step forward, a 25 percent chance she'll stay in place and a 50 percent chance she'll take one step back toward the couch. In the long run, what percent of the time does the baby choose to clutch the couch?

8: Louie lives in a town that had a 50 percent chance of rain each morning and an independent 40 percent chance each evening. Louie walks to and from work each day, bringing an umbrella with him if it was raining and not bringing one if it wasn't. On Sunday night, two of his three umbrellas were with him at home and one was at his office. What were the chances that he made it through the work week without getting wet?

9: You're going to play a game. Like many probability games, this one involves an infinite supply of ping-pong balls.

The balls are numbered 1 through N. There is also a group of N cups, labeled 1 through N, each of which can hold an unlimited number of ping-pong balls. The game is played in rounds. A round is composed of two phases: throwing and pruning.

During the throwing phase, the player takes balls randomly, one at a time, from the infinite supply and tosses them at the cups. The throwing phase is over when every cup contains at least one ping-pong ball. Next comes the pruning phase. During this phase the player goes through all the balls in each cup and removes any ball whose number does not match the containing cup.

Every ball drawn has a uniformly random number, every ball lands in a uniformly random cup, and every throw lands in some cup. The game is over when, after a round is completed, there are no empty cups.

(a) How many balls would you expect to need to draw and throw to finish this game?

(b) For $N = 3$, how many rounds would you expect to need to play to finish this game?

Books

References

[1] Richard Durrett. *Elementary Probability for Applications*. Cambridge University Press, 2009.

[2] Charles M. Grinstead and J. Laurie Snell. *Introduction to Probability*. American Mathematical Society, 1997.

[3] J. G. Kemeny, J. L. Snell, and G. L. Thompson. *Introduction to Finite Mathematics*. Prentice-Hall, 1974.

[4] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. D. Van Nostrand Company, Inc., 1960.

[5] David A. Levin, Yuval Peres, and Elizabeth L. Wilmer. *Markov Chains and Mixing Times*. American Mathematical Society, 2009.

[6] Sheldon M. Ross. *Introduction to Probability Models*. Academic Press, 9th edition, 2006.

[7] David Stirzaker. *Stochastic Processes and Models*. Oxford University Press, 2005.

Links

1. *Introduction to Probability*, Grinstead and Snell. Chapter 11 - Markov Chains http://tinyurl.com/qw6sa Introduction to Probability, Chapter 11 Accessed Wed Jan 16 07:32:50 CST 2019

2. Markov The markovchain package Accessed Wed Jan 16 07:34:15 CST 2019

3. Getting Started with Markov Chains Accessed Wed Jan 16 07:32:22 CST 2019

Solutions

1: Problem created by LT Grant, 2006.

(a) There are 4 states, D for dropping out, 1 for first-year, 2 for second-year, and G for graduating.

$$P = \begin{array}{c} \\ D \\ 1 \\ 2 \\ G \end{array} \begin{array}{c} \begin{array}{cccc} D & 1 & 2 & G \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 3/20 & 1/4 & 3/5 & 0 \\ 1/10 & 0 & 1/5 & 7/10 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}.$$

(b) Absorbing states are D and G. Transient states are 1, 2.

(c)

$$P^2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 99/400 & 1/16 & 27/100 & 21/50 \\ 31/250 & 0 & 1/125 & 217/250 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Probability of graduating on time is 21/50.

$$P^3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2271/8000 & 1/64 & 183/2000 & 609/1000 \\ 31/250 & 0 & 1/125 & 217/250 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

(d) Probability of graduating in 3 years is 609/1000.

(e) Canonical form is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 3/20 & 0 & 1/4 & 3/5 \\ 1/10 & 7/10 & 0 & 1/5 \end{pmatrix}$$

so

$$N = \begin{pmatrix} 4/3 & 1 \\ 0 & 7/8 \end{pmatrix} \qquad B = NR = \begin{pmatrix} 3/10 & 7/10 \\ 1/8 & 7/8 \end{pmatrix}.$$

The chance that a student will graduate eventually is 7/10. The chance the student will drop out eventually is 3/10.

(f) The expected time to graduation is $4/3 + 1 = 7/3$ years.

2: Problem created by LT Grant, 2006.

(a)

$$P = \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \\ F \end{array} \begin{array}{c} \begin{array}{cccccc} A & B & C & D & E & F \end{array} \\ \begin{pmatrix} 0 & 1/3 & 1/3 & 0 & 1/3 & 0 \\ 1/2 & 0 & 0 & 1/2 & 0 & 0 \\ 1/2 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 1/3 & 1/3 & 0 & 0 & 1/3 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{array}.$$

(b) Absorbing states are E and F. Transient states are A, B, C, D.

(c)

$$P' = \begin{array}{c} \\ E \\ F \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccccc} E & F & A & B & C & D \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1/3 & 0 & 0 & 1/3 & 1/3 & 0 \\ 0 & 0 & 1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 1/2 & 0 & 0 & 1/2 \\ 0 & 1/3 & 0 & 1/3 & 1/3 & 0 \end{pmatrix} \end{array}.$$

$$N = \begin{pmatrix} 2 & 1 & 1 & 1 \\ \frac{3}{2} & 2 & 1 & \frac{3}{2} \\ \frac{3}{2} & 1 & 2 & \frac{3}{2} \\ 1 & 1 & 1 & 2 \end{pmatrix}.$$

(d) $(N\mathbf{1})_1 = 2 + 1 + 1 + 1 = 5$.

(e) From A, probability to E is $2/3$, to F is $1/3$. From D, probability to E is $1/3$, to F is $2/3$. Seems reasonable by symmetry.

3: Problem created by LT Grant, 2006.

(a) No, expect absorption probabilities to increase.

(b) From A, probability to E to increase, to F to decrease. From D, probability to E to increase, to F to decrease.

(c)

$$\begin{pmatrix} 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ \frac{1}{3} & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

(d) From A, probability to E is 0.792, to F to decrease is 0.208. From D, probability to E is approximately 0.458, to F is approximately 0.542.

4: Adapted from Karlin and Taylor, *A First Course in Stochastic Processes*, second edition, page 73, problem 1(b).

$$P_{jk} = \begin{cases} (\frac{j}{N})^2 & k = j-1, \quad j = 1, 2, \ldots, N \\ (2\frac{j}{N})\frac{N-j}{N} & k = j, \quad j = 0, 1, 2, \ldots, N \\ 1 - (\frac{j}{N})^2 & k = j-1, \quad j = 0, 1, 2, \ldots, N-1 \end{cases}$$

There are no absorbing states, all states are recurrent. Yes ergodic. As an example, for $N = 3$, the transition probability matrix is

$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1/9 & 4/9 & 4/9 & 0 \\ 0 & 4/9 & 4/9 & 1/9 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

with stationary distribution $\pi = (1/20, 9/20, 9/20, 1/20)$ For $N = 4$, the transition probability matrix is

$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1/16 & 6/16 & 9/16 & 0 & 0 \\ 0 & 4/16 & 8/16 & 4/16 & 0 \\ 0 & 0 & 9/26 & 6/16 & 1/16 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

5: Adapted from Karlin and Taylor, *A First Course in Stochastic Processes*, second edition, page 74, problem 2(a).

$$P_{jk} \begin{cases} \left(\frac{j}{N}\right) q \qquad k = j-1, \qquad j = 1, 2, \ldots, N \\ \left(\frac{j}{N}\right) p + \left(\frac{N-j}{N}\right) q \qquad k = j, \qquad j = 0, 1, 2, \ldots, N \\ \left(1 - \frac{j}{N}\right) p \qquad k = j+1, \qquad j = 0, 1, 2, \ldots, N-1 \end{cases}$$

An example in the case $N = 4$

$$\begin{pmatrix} q & p & 0 & 0 & 0 \\ \frac{q}{4} & \frac{3q}{4} + \frac{p}{4} & \frac{3p}{4} & 0 & 0 \\ 0 & \frac{q}{2} & \frac{q}{2} + \frac{p}{2} & \frac{p}{2} & 0 \\ 0 & 0 & \frac{3q}{4} & \frac{q}{4} + \frac{3p}{4} & \frac{p}{4} \\ 0 & 0 & 0 & q & p \end{pmatrix}.$$

The unnormalized stationary distribution is $[1, 4p/q, 6p^2/q^2, 4p^3/q^3, p^4/q^4]$, for $p = 1/2 = q$, normalized, this is $[1/16, 1/4, 3/8, 1/4, 1/16]$, for $p = 3/4$, $q = 1/4$, normalized, this is approximately $[0.007, 0.082, 0.370, 0.740, 0.555]$.

6: Adapted from Karlin and Taylor, *A First Course in Stochastic Processes*, second edition, page 74, problem 2(b).

$$P_{jk} = \begin{cases} \left(1 - \frac{j}{N}\right) p \qquad k = j-1, \qquad j = 1, 2, \ldots, N \\ \left(\frac{j}{N}\right) p + \left(\frac{N-j}{N}\right) q \qquad k = j, \qquad j = 0, 1, 2, \ldots, N \\ \left(\frac{j}{N}\right) q \qquad k = j+1, \qquad j = 0, 1, 2, \ldots, N-1. \end{cases}$$

7: Adapted from FiveThirtyEight Riddler who in turn got the problem from Steve Simon. From numerical experiments with the transition probability matrix, about 50% of the time. This matrix converges fairly slowly to the limiting matrix, so experiments require high powers.

8: Adapted from FiveThirtEight Riddler who in turn got the problem from Josh Vandenham.)

Remark. q This problem and the solution as a Markov Chain appeared in FiveThirtyEight Riddler column of December 14, 2018.

Just keep track of the different "states" of Louie's umbrellas. He could have three at home and zero at work, two at home and one at work, one at home and two at work, or zero at home and three at work. The movement through the random states (the work week) is described by the transition

matrix, which contains the probabilities of umbrellas moving from one place to another on any given day. In our particular case, the matrix looks like this:

$$\begin{pmatrix} 0.3 & 0.2 & 0.0 & 0.0 & 0.5 \\ 0.3 & 0.5 & 0.2 & 0.0 & 0.0 \\ 0.0 & 0.3 & 0.5 & 0.2 & 0.0 \\ 0.0 & 0.0 & 0.3 & 0.5 & 0.2 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{pmatrix}.$$

In this solution, there are five states: zero, one, two or three umbrellas at home plus the state in which Louie has already gotten wet. The rows of the matrix are Louie's current state, and the columns are his destination. For example, he starts in the third row (two umbrellas at home). There is no chance he'll end up with zero umbrellas at home, there's a 0.3 chance he'll end the day with just one at home, a 0.5 chance with two at home, a 0.2 chance with three at home, and no chance he'll get wet on this first day. To get the probabilities for the entire work week, simply multiply this matrix by itself five times, representing the meteorological randomness unfolding over the five days, which gives us the following:

$$\begin{pmatrix} 0.04791 & 0.07634 & 0.04976 & 0.01776 & 0.80823 \\ 0.11451 & 0.19889 & 0.15274 & 0.06752 & 0.46634 \\ 0.11196 & 0.22911 & 0.22553 & 0.12610 & 0.3073 \\ 0.05994 & 0.15192 & 0.18915 & 0.12425 & 0.47474 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Last, look at the third row (the state in which he began) and the final column: That's the chance he gets wet during the week. Subtract from 1 to find the chance that he didn't get wet and get 69.27 percent.

Remark. This is a version of Ross, page 181, problem number 22

9:

Remark. This problem and the solution as a Markov Chain appeared in FiveThirtyEight Riddler column of November 16, 2018. The solution is adapted from Five Thirty Eight Riddler.

(a) For this question, the rounds don't matter at all. The game will end once there is at least one correct ball in each cup, and the pruning phase at the end of each round only removes incorrect balls so effectively neglect it.

This is a classical coupon collector problem. Define Y_i, for i from 0 to $N-1$, to be the number of additional tosses that need to be obtained after i distinct cups have been filled. Define Y as the total number of tosses required to fill the cups with correspondingly numbered balls so

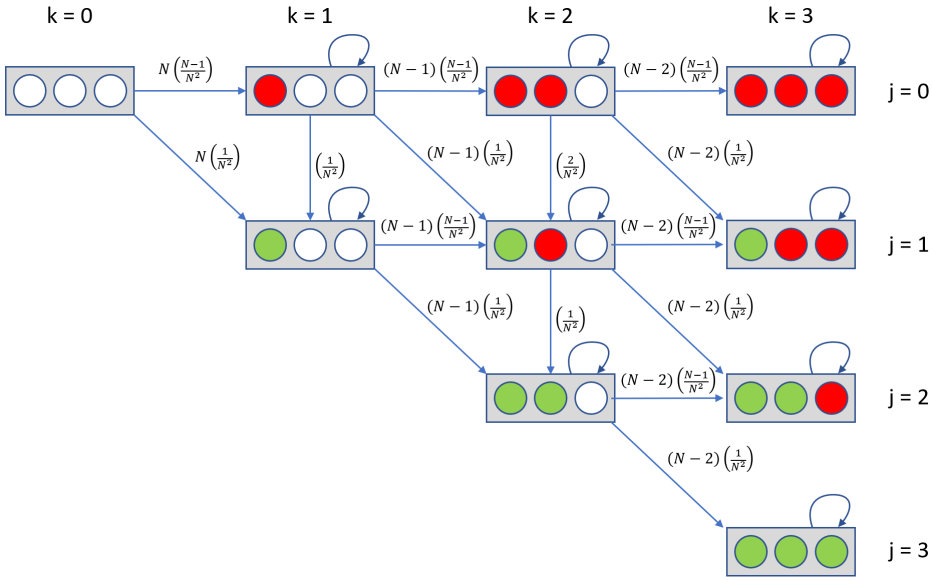
$$Y = Y_0 + Y_1 + Y_2 + \cdots + Y_{N-1}.$$

When i distinct cups have been filled with corresponding numbered balls, the probability that a ball with a number of an unfilled cup is drawn is $\frac{N-i}{N}$. Each time a ball is thrown, there is a $1/N$ chance it will land in the correct cup. It follows that the probability a new cup is filled with the correctly numbered ball is $\frac{N-i}{N} \cdot \frac{1}{N}$. Thus, Y_i is a geometric random variable with success parameter $\frac{N-i}{N^2}$. Then $\mathbb{E}[Y_i] = \frac{N^2}{N-i}$. Therefore

$$\begin{aligned}
\mathbb{E}[Y] &= \mathbb{E}[Y_0] + \mathbb{E}[Y_1] + \mathbb{E}[Y_2] + \cdots + Y_{N-1} \\
&= N + \frac{N^2}{N-1} + \frac{N^2}{N-2} + \cdots + \frac{N^2}{1} \\
&= N^2 \sum_{k=1}^{N} \frac{1}{k} \approx N^2(\log N + \gamma)
\end{aligned}$$

where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant.

(b) This is a more complicated question, so to illustrate the approach solve it for the case $N = 3$. Begin by breaking each round into a number of stages, which track the progress of cups as they are filled. In the diagram of a single below, each cup is either empty (no ball), is red (incorrect ball), or is green (correct ball). The transition probabilities are on the arrows.

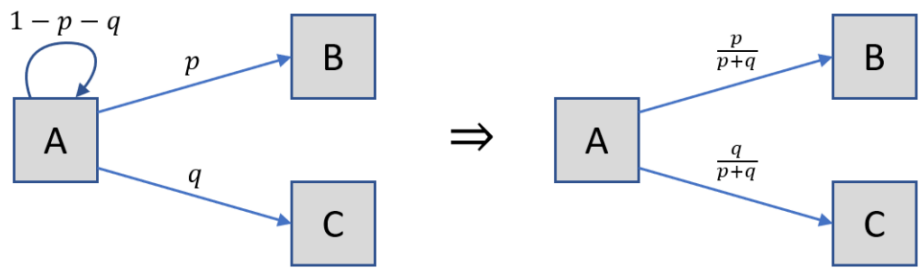


The integer k indicates the number of balls thrown. The integer j indicates the number of cups filled holding a correspondingly numbered ball. A colored cup may contain more than one ball. The cup position left to right is not associated with the cup or ball numbers, the position merely indicates the number of correctly or incorrectly filled cups. For example, if 2 balls have been thrown into 2 cups, one correctly numbered, one incorrectly numbered, is the center diagram. Then on the next ball thrown, that ball can incorrectly go into the empty cup with probability $\frac{N-i}{N} \cdot \frac{1}{N}$. The correct ball can go into the correct cup with probability $\frac{1}{N2}$. In this diagram, self-loops have whatever probability is required so that the outgoing arrows sum to 1. This is a Markov chain. We start on the initial state on the left (all cups empty), and advance to the absorbing state on the right (all cups full). The transition matrix for $N = 3$, arranging the states from top to bottom and left to right,

is:

$$A = \begin{pmatrix} 0 & \frac{2}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2}{9} & \frac{1}{9} & \frac{4}{9} & \frac{2}{9} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 & \frac{4}{9} & \frac{2}{9} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{4}{9} & \frac{2}{9} & 0 & \frac{2}{9} & \frac{1}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5}{9} & \frac{1}{9} & 0 & \frac{2}{9} & \frac{1}{9} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2}{3} & 0 & 0 & \frac{2}{9} & \frac{1}{9} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

First, eliminate the self-loops by normalizing because it doesn't matter how much time is spent at each stage, the state of the Markov chain hasn't changed. See below for an illustration of this normalization, based on the observation that $\mathbb{P}[B \mid A] = \mathbb{P}[B \mid B \cup C] = \frac{p}{p+q}$.



Here is what our transition matrix looks like after normalization.

$$A_{\text{norm}} = \begin{pmatrix} 0 & \frac{2}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{7} & \frac{4}{7} & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2}{5} & 0 & \frac{2}{5} & \frac{1}{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & \frac{1}{2} & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Once again, the rows sum to 1, but this time only the absorbing states

have self-loops. Putting the matrix into canonical form:

$$A_{\text{canon}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2}{3} & \frac{1}{3} & 0 & 0 & \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & \frac{4}{7} & \frac{2}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & \frac{2}{3} & \frac{1}{3} & 0 \\ \frac{2}{5} & \frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{5} & 0 \\ 0 & \frac{1}{2} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & \frac{2}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The normal matrix $(I = Q)^{-1}$ is then

$$\begin{pmatrix} 1 & \frac{2}{3} & \frac{3}{7} & \frac{2}{3} & \frac{7}{9} & \frac{7}{36} \\ 0 & 1 & \frac{1}{7} & \frac{2}{3} & \frac{7}{9} & \frac{7}{36} \\ 0 & 0 & 1 & \frac{2}{3} & \frac{7}{9} & \frac{7}{36} \\ 0 & 0 & 0 & 1 & \frac{2}{3} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & 1 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then the expected number of throws until the cups are filled is (as a column vector)

$$\left(\frac{941}{252}, \frac{701}{252}, \frac{95}{36}, \frac{11}{6}, \frac{5}{4}, 1\right)^T.$$

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